# Ngspice quick guide

Ngspice is an open-source circuit simulator that has a lot of basic and advanced functions, which is well-suited for academic research.

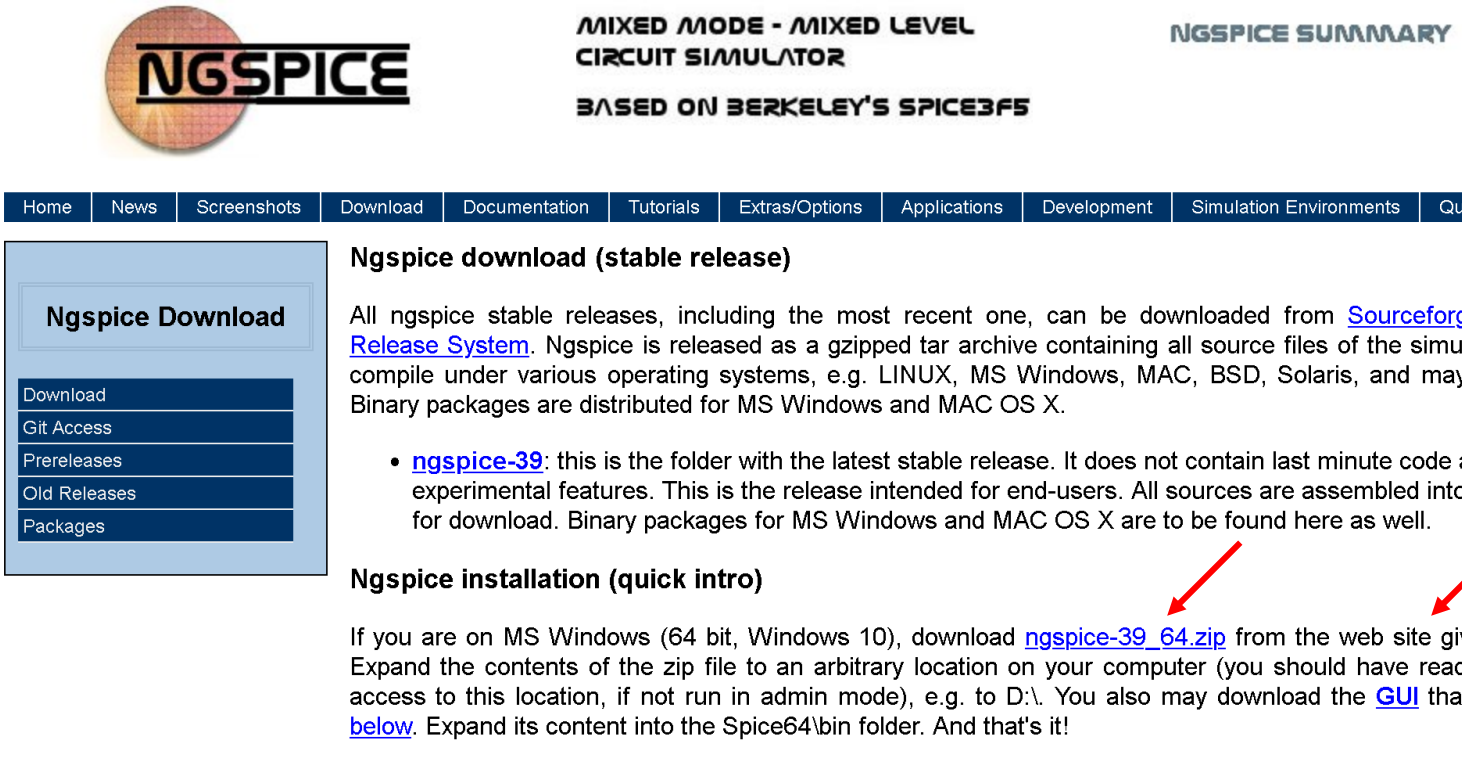
Download ngspice-39_64.zip and the GUI at

http://ngspice.sourceforge.net/download.html

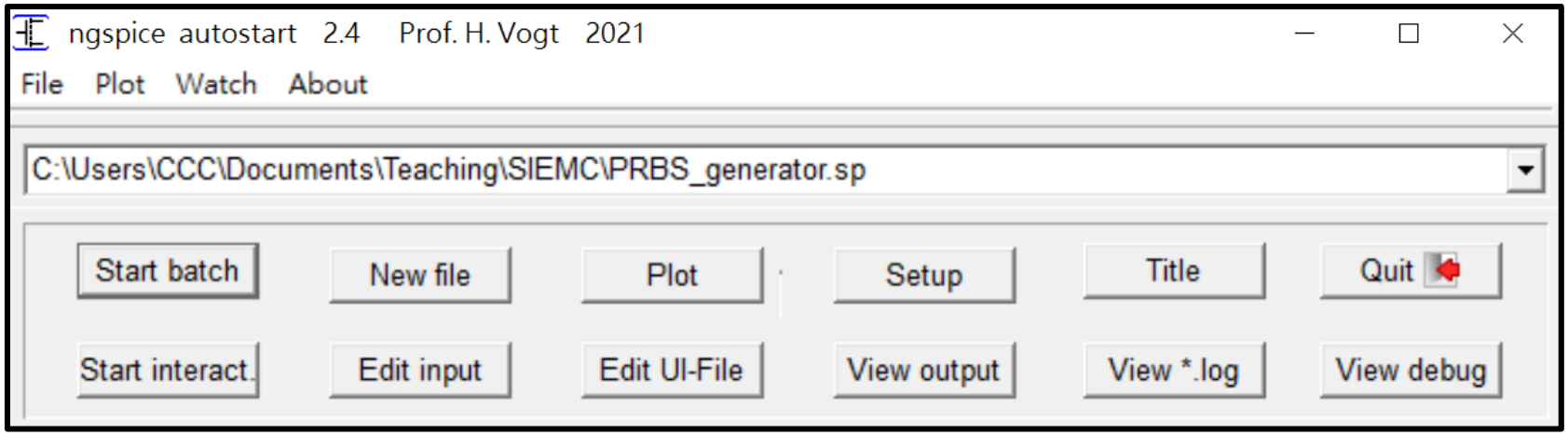
Expand the contents of ngspice-39_64.zip into an arbitrary folder.

Expand the contents of the GUI into the Spice64\bin folder.

NOTE: Make sure the whole path names of ngspice.exe and the circuit files do not have any whitespace or Chinese characters. Otherwise, there may be an error.



Open DuSpiceStart.exe. The first time you open, it may ask you to choose a spice. Select ngspice.exe.




Click “New file”. Choose your circuit file.

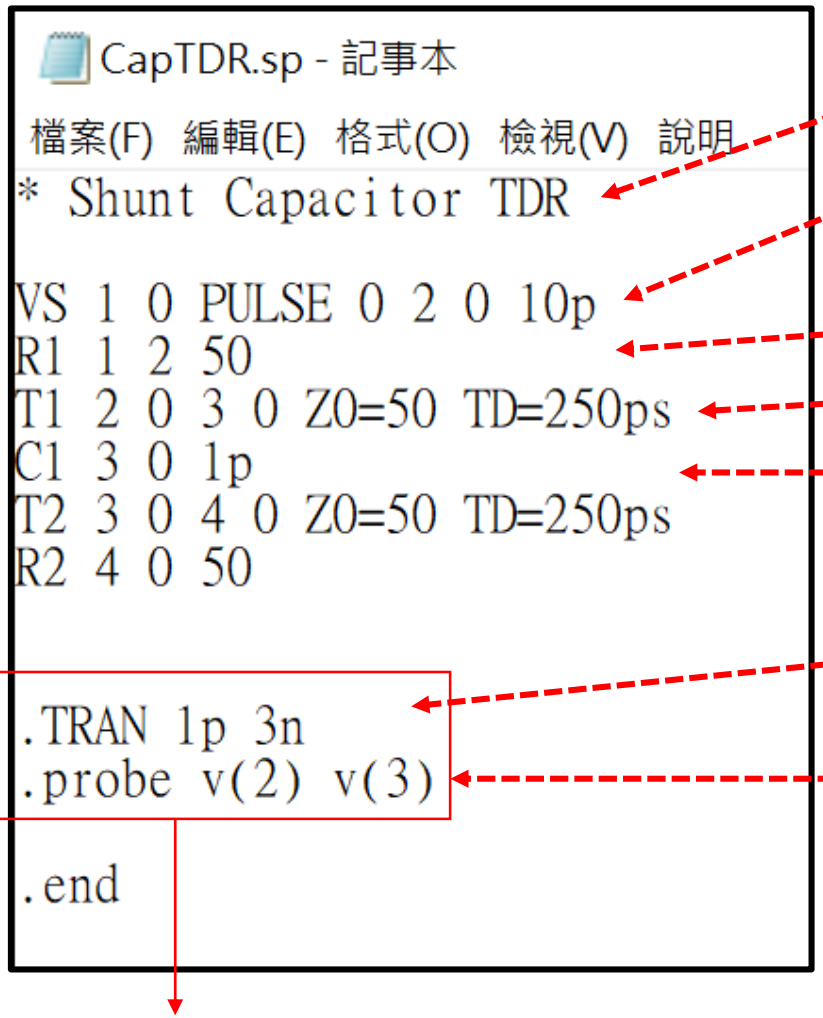
Click “Start batch”, which means run simulation.

Depending on your circuit file, there are two possible ways to process the data.

Method 1: click “Plot” to access the data

Method 2: save the data to a text file and use Excel/Python to read it.

Circuit file example: The TDR waveform for shunt capacitor




1st line is always comment

Voltage source. Pulse function: 0-to-2 volt, delay=0, trise=10ps.

Resistor 50Ω

Ideal transmission line

Capacitor 1pF

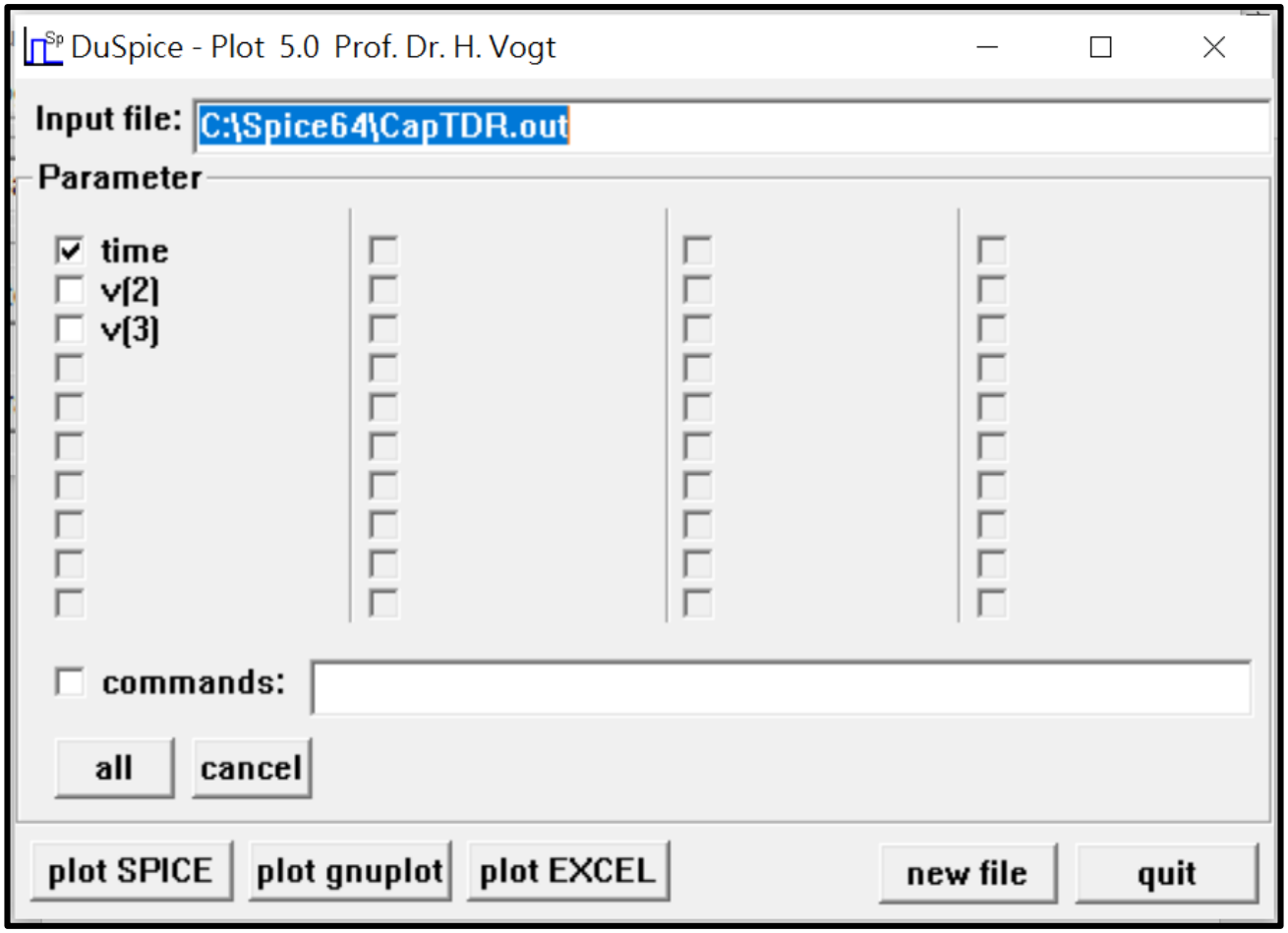
Transient simulation, timestep=1ps, stoptime=3ns

Only save the voltages at node 2 and node 3. All other voltages and currents will be discarded after simulation.

If the commands in the circuit file are in this form, then use Method 1 to access the data.

For detailed instruction on the rules for input netlist, see Ngspice User's Manual at Spice64\docs.

Click “Plot” and you will see this window




Choose the variable(s) you wish to plot, or enter the expression at “commands”.
To quickly view the waveforms, click “plot SPICE”.
To further edit the curves, and combine with other curves, click “plot EXCEL”. An excel file containing the data and plot will be automatically created.

Method 1 of accessing the data is convenient. However, the maximum number of data that can be exported to Excel is limited. So for large simulation, we can use method 2.

```
* PRBS generator (pseudo random bit sequence)

* define unit interval (= 1 / bitrate)
.param UI = 100p

* trrandom = time domain random voltage
* 1st argument: random number type (2=normal)
* 2nd argument: duration of each draw
* 3rd argument: time delay before start
Vrand 1 0 trrandom (2 UI UI)
R1 1 0 1

* random square wave, rise time~10ps, 0~1 V
Bprbs 2 0 v = v(1) < 0 ? 0 : 1
Rload 2 3 50
Cload 3 0 0.1p

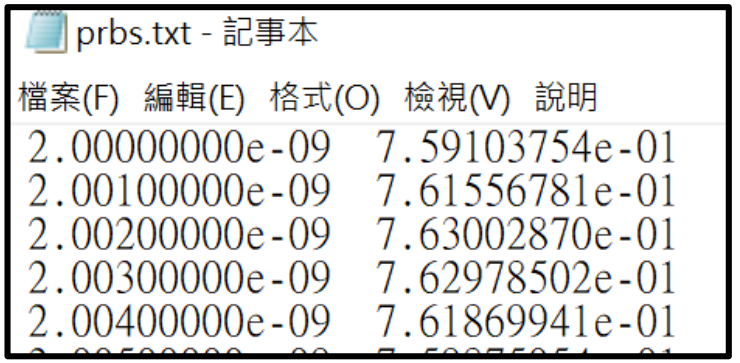
.control
tran 1p 1n
*option interp
linearize v(3)
set wr_singlescale
wrdata prbs.txt v(3)
.endc

.end
```

First, put all the commands in the circuit file inside a .control … .endc block, and include the last two lines,

set wr_singlescale
wrdata [filename] [variable name]

Then the output data will be written to a text file located at the ngspice.exe directory (Spice64\bin).




This type of file can be easily read into Excel or Python, e.g. by numpy.loadtxt.

# Run Ngspice in Python (感謝林婷萱同學整理)

## Step 1: Add ngspice.exe into system path.

指令:

參考 https://zhuanlan.zhihu.com/p/174091151

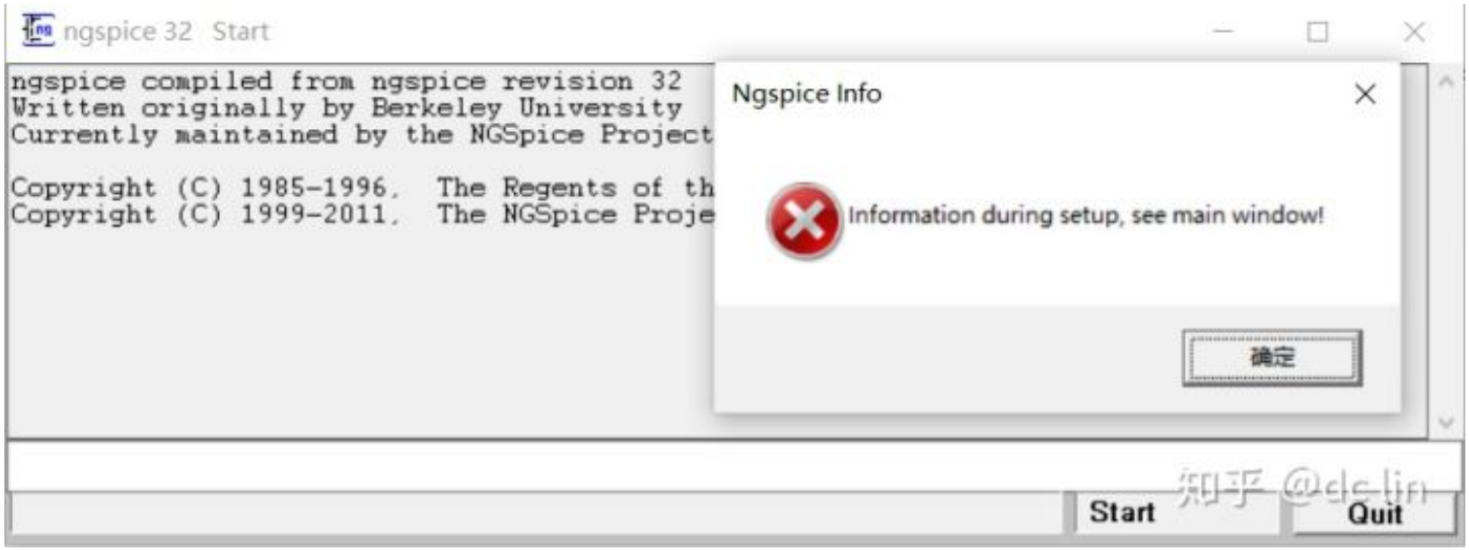
Windows下，輸入：<win> + <r>組合鍵，輸入cmd換行，在彈出的命令窗口中，輸入：

```
setx path "D:\Spice64\bin;%path%"
```

將ngspice執行程序ngspice.exe的路徑添加到系統的搜索路徑中。

要驗證成功，請打開任何命令行窗口，輸入任何目錄，然後鍵入：

```
ngspice -v
```



Step 2: In Python (Spyder), run the following code
The circuit file needs to be at the current working directory (not necessarily the directory of ngspice.exe) of Python. A log.txt file, in addition to any saved data file, will be generated at the same directory after simulation.

```
import subprocess
subprocess.run("ngspice -o log.txt -b PRBS_generator_2.sp")
```

